

ASHLEY (YE) GAO

Curriculum Vitae

Assistant Professor, Department of Computer Science, William & Mary
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I. EDUCATION

Ph.D., Computer Science , University of Virginia	2019–2023
Advisor: John A. Stankovic. Dissertation: “Addressing Realisms Faced by Deep Learning Models in Cyber Physical Systems.”	
M.S., Computer Science , University of Virginia	2017–2019
B.S., Computer Science , University of California, San Diego	2012–2017
B.A., Literatures of the World , University of California, San Diego	2012–2017

II. ACADEMIC APPOINTMENTS

Assistant Professor , Department of Computer Science, William & Mary	2023–present
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III. AREAS OF RESEARCH

My research lies at the intersection of affective computing, audio-language modeling, and robust machine learning. I develop methods for emotion-aware representation learning and generation, with a focus on audio large language models, controllable text-to-speech, and speech emotion recognition.

Current Research Directions

- D3. Domain-generalized audio sensing — robust speech emotion recognition under distribution shift
- D2. Sentiment steering in large language models — controllable generation of emotional tone
- D1. Audio stress detection for ADRD caregivers — mobile and scalable systems for caregiver support

IV. SCHOLARSHIP

Names marked with † are graduate students advised by the author: Jiacheng Shi (†) and Hongfei Du (†). Collin Macdonald (†) was advised by the author when the paper was published.

A. Refereed Conference and Workshop Proceedings

- C18. Du, H.†, Shi, J.†, Zhang, Y., & **Gao, Y.** (2026). Tracing a sparse emotion-control circuit in LLM-based text-to-speech. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*.
- C17. Du, H.†, Shi, J.†, Zhang, Y., Zhou, G., & **Gao, Y.** (2026). Usage-modulated sentiment representations in large language models. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*.
- C16. Du, H.†, Shi, J.†, Lu, S., Zhou, G., & **Gao, Y.** (2026). Sparse autoencoders for interpretable emotion control in text-to-speech. In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*, *Proceedings of Machine Learning Research (PMLR)*.
- C15. Shi, J.†, Du, H.†, Song, X., Hong, Y. A., Zhang, Y., & **Gao, Y.** (2026). Emo-BPO: Emotion bidirectional preference optimization for diffusion-based emotional TTS. In *Proceedings of Interspeech 2026*, Sydney, Australia, September 28–October 1, 2026, ISCA. (Forthcoming.)
- C14. Shi, J.†, Du, H.†, Song, X., Hong, Y. A., Zhang, Y., & **Gao, Y.** (2026). AffectCodec: Emotion-preserving neural speech codec for expressive speech modeling. In *Findings of the Association for Computational Linguistics: ACL 2026*, San Diego, California, July 2026, Association for Computational Linguistics, pp. 9102–9124.
- C13. Nair, T., Nafee, M. W., Jiang, M., **Gao, Y.**, Chen, H., & Zhang, Y. (2026). Confidence-aware ranker ensembles for robust in-context knowledge editing. In *Findings of the Association for Computational*

Linguistics: ACL 2026, San Diego, California, July 2026, Association for Computational Linguistics, pp. 35070–35080.

- C12. Shi, J.†, Du, H.†, He, Y., Hong, Y. A., & **Gao, Y.** (2026). Emotion-aligned generation in diffusion text-to-speech models via preference-guided optimization. In *ICASSP 2026 — 2026 IEEE International Conference on Acoustics, Speech and Signal Processing*, Barcelona, Spain, May 4–8, 2026, IEEE.
- C11. Shi, J.†, Du, H.†, Hong, Y. A., & **Gao, Y.** (2026). EMO-TTA: Improving test-time adaptation of audio-language models for speech emotion recognition. In *ICASSP 2026 — 2026 IEEE International Conference on Acoustics, Speech and Signal Processing*, Barcelona, Spain, May 4–8, 2026, IEEE.
- C10. Shi, J.†, Du, H.†, Hong, Y. A., & **Gao, Y.** (2026). Plug-and-play emotion graphs for compositional prompting in zero-shot speech emotion recognition. In *ICASSP 2026 — 2026 IEEE International Conference on Acoustics, Speech and Signal Processing*, Barcelona, Spain, May 4–8, 2026, IEEE.
- C9. Mersa, O., Jiang, M., **Gao, Y.**, Li, Q., & Zhang, Y. (2026). Hierarchical convolution multibranch transformer for EEG signals. In *ICASSP 2026 — 2026 IEEE International Conference on Acoustics, Speech and Signal Processing*, Barcelona, Spain, May 4–8, 2026, IEEE.
- C8. Jiang, M., Chen, Y., Liu, G., Sahu, A., **Gao, Y.**, & Zhang, Y. (2026). Pairs, not labels: Predicting protein-phenotype associations via link prediction. In *ICASSP 2026 — 2026 IEEE International Conference on Acoustics, Speech and Signal Processing*, Barcelona, Spain, May 4–8, 2026, IEEE.
- C7. Du, H.†, Shi, J.†, Myerston, J., Lu, S., Zhou, G., & **Gao, Y.** (2025). Role-guided annotation and prototype-aligned representation learning for historical literature sentiment classification. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, Suzhou, China, November 2025, Association for Computational Linguistics, pp. 3756–3768.
- C6. Du, H.†, Lu, S., Zhou, G., & **Gao, Y.** (2025). EAA: Emotion-aware audio large language models with dual cross-attention and context-aware instruction tuning. In *Proceedings of Interspeech 2025*, Rotterdam, The Netherlands, August 2025, ISCA, pp. 5433–5437.
- C5. Shi, J.†, Zhang, Y., & **Gao, Y.** (2025). CLEP-DG: Contrastive learning for speech emotion domain generalization via soft prompt tuning. In *Proceedings of Interspeech 2025*, Rotterdam, The Netherlands, August 2025, ISCA, pp. 4498–4502.
- C4. Clapham, J., Zhou, M., MacDonald, C., Koltermann, K., **Gao, Y.**, & Shao, H. (2025). ElectroMeter: The practical electrolyte measurement system. In *Proceedings of the ACM/IEEE International Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE 2025)*, ACM.
- C3. Koltermann, K., Clapham, J., Blackwell, G., Jung, W., Burnet, E. N., **Gao, Y.**, Shao, H., Cloud, L., Pretzer-Aboff, I., & Zhou, G. (2024). Gait-Guard: Turn-aware freezing of gait detection for non-intrusive intervention systems. In *2024 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, Wilmington, DE, June 2024, IEEE, pp. 61–72. **Best Paper Award.**
- C2. **Gao, Y.**, Baucom, B., Gordon, K. C., Rose, K. M., Wang, H., & Stankovic, J. A. (2023). E-ADDA: Unsupervised adversarial domain adaptation enhanced by a new Mahalanobis distance loss for smart computing. In *2023 IEEE International Conference on Smart Computing (SMARTCOMP)*, IEEE, pp. 172–179. **Best Paper Runner-Up.**
- C1. **Gao, Y.**, Ma, M., Gordon, K. C., Rose, K. M., Wang, H., & Stankovic, J. A. (2020). A monitoring, modeling, and interactive recommendation system for in-home caregivers: Demo abstract. In *Proceedings of the 18th ACM Conference on Embedded Networked Sensor Systems (SenSys '20)*, ACM, pp. 587–588.

B. Refereed Journal Articles

- J5. Ko, E., **Gao, Y.**, Wang, P., Wijayasingha, L., Wright, K. D., Gordon, K. C., Wang, H., Stankovic, J. A., & Rose, K. M. (2025). Challenges to recruiting dementia caregiving dyads in community-based settings: Lessons learned from an in-home technology-based intervention. *JMIR Formative Research*, 9(1), e59291.
- J4. Macdonald, C†., Chu, Z., Stankovic, J. A., Zhou, G., Shao, H., & **Gao, Y.** (2024). MiddleGAN: Generating inter-domain images with GANs. *Transactions on Machine Learning Research (TMLR)*, 2024. (Open-access journal; articles are not assigned page numbers.)

- J3. **Gao, Y.**, Salekin, A., Gordon, K. C., Rose, K. M., Wang, H., & Stankovic, J. A. (2021). Emotion recognition robust to indoor environmental distortions and non-targeted emotions using out-of-distribution detection. *ACM Transactions on Computing for Healthcare (HEALTH)*, 3(2), 1–22.
- J2. **Gao, Y.**, Jabbour, J., Schlegel, E. C., Ma, M., McCall, M., Wijayasingha, L., Ko, E., Rose, K. M., Wang, H., & Stankovic, J. A. (2021). Out-of-the-box deployment to support research on in-home care of Alzheimer’s patients. *IEEE Pervasive Computing*, 21(1), 37–47.
- J1. Rose, K. M., Coop Gordon, K., Schlegel, E. C., McCall, M., **Gao, Y.**, Ma, M., Wijayasingha, L., Stankovic, J. A., & Wang, H. (2021). Smarthealth technology study protocol to improve relationships between older adults with dementia and family caregivers. *Journal of Advanced Nursing*, 77(5), 2519–2529.

C. Published Creative Writing

- W2. **Gao, A.** (2026). *The Tempest’s Call*. Pegasus Publishers, United Kingdom. (Forthcoming.)
- W1. **Gao, A.** (2026). The purpose of a name. *Neon Dystopia*, United States. Short fiction.

D. Grants and Contracts

- G1. Agentic AI for Multi-Modal Perception Integrity: Threat Reasoning and Response in Cyber-Physical Transportation Systems.
- Commonwealth Cyber Initiative, Coastal Virginia (COVA CCI), PI: Qing Tang of Old Dominion University. Overall award: \$80,000. Author’s Role: **Co-Principal Investigator**. Allocated funds to W&M: \$39,199. 2026.

V. TEACHING AND MENTORING

A. Courses Taught — Graduate

All courses listed below were taught at William & Mary.

CSCI 680: Affective Computing	2026 F
CSCI 680: Deep Transfer Learning	2025 F/S
CSCI 516: Fundamentals of Artificial Intelligence and Machine Learning	2023 F, 2024 F, 2026 S

B. Courses Taught — Undergraduate

CSCI 416: Fundamentals of Artificial Intelligence and Machine Learning	2023 F, 2024 F, 2026 S
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C. Doctoral Students Advised (The Galatea Lab)

Hongfei Du (†), Ph.D. student, William & Mary	2024–present
First-author publications: ICML, EMNLP (Findings), Interspeech.	
Jiacheng Shi (†), Ph.D. student, William & Mary	2024–present
First-author publications: ICASSP, ACL (Findings), Interspeech.	
Hyo Jin Jon (†), Ph.D. student, William & Mary	2026–present
Siyuan Zhao (†), Ph.D. student, William & Mary	2026–present

D. Undergraduate Students Advised

Ryan Hong , University of Maryland	2026–present
Meave Meyer , William & Mary	2026–present
Lilly Liu , William & Mary — initial placement: Carnegie Mellon University	2024–2026

VI. SERVICE

A. Editorial Positions

Associate Editor, <i>Elsevier Smart Health</i>	2024–present
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B. Grant Review Panels (Invited)

Panelist, National Institutes of Health (NIH)	2025
Panelist, National Institutes of Health (NIH)	2024
Panelist, National Science Foundation (NSF)	2024

C. Conference Organization

Demo & Poster Chair, IEEE/ACM Conference on Connected Health (CHASE)	2024
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D. Reviewing

<i>IEEE Affective Computing</i>	2026
ICML; AAAI; ICLR	2025
NeurIPS; AAAI; ICLR; <i>ACM Transactions on Computing for Healthcare</i>	2024
AAAI; NeurIPS; ICLR	2023
<i>IEEE Affective Computing</i> ; AAAI; <i>ACM IMWUT/UbiComp</i> ; NeurIPS	2022
<i>ACM Health</i>	2021

E. Professional Memberships

Association for Computing Machinery (ACM)	2025
Institute of Electrical and Electronics Engineers (IEEE)	2025

VII. HONORS AND AWARDS

A. Research and Service Awards

Best Paper Award, IEEE/ACM CHASE	2024
Best Service Award, IEEE/ACM CHASE	2024
Best Paper Runner-Up Award, IEEE SmartComp	2023
Best Poster Award, Computer Science Symposium, University of Virginia	2022

B. Fellowships and Academic Honors

Ph.D. Fellowship, Department of Computer Science, University of Virginia	2019
Academic Excellence Fellowship, Department of Computer Science, University of Virginia	2017
Honors with High Distinction, University of California, San Diego	2016

VIII. MEDIA AND RECOGNITION

Featured as Honors Student of the Department of Literature, University of California, San Diego	2016
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